2023 2nd International Conference on Machine Learning, Cloud Computing and Intelligent Mining (MLCCIM) | 979-8-3503-2853-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/MLCCIM60412.2023.00083

Evaluation of landslide susceptibility in Helan Mountain by fusion of cluster analysis and remote sensing images

1st Tianke Fang
School of Computer and Information Engineering,
Xiamen University of Technology, 361024, Fujian
Xiamen, China
rock_ftk@163.com

2nd Zhenxing Hui*
School of Computer and Information Engineering,
Xiamen University of Technology, 361024, Fujian
Xiamen, China
779551378@qq.com

3rd Liheng Zhang
School of Computer and Information Engineering,
Xiamen University of Technology, 361024, Fujian
Xiamen, China
2824407943@qq.com

***Abstract*—In order to assess the degree of influence of each landslide influence factor on landslides, this paper takes Helan Mountain as the research object, and collects data from 56 historical landslide sites in Helan Mountain through field surveys and remote sensing images. Through cluster analysis, the landslide influencing factors were categorized into eight main major categories, including slope, slope direction, curvature, planar curvature, profile curvature, terrain undulation, surface roughness, and terrain humidity index. Then, a comprehensive evaluation was carried out using the principal component analysis integrated evaluation method, and the evaluation results showed that terrain undulation and surface roughness had high indices in the evaluation, while the curvature and terrain humidity indices were relatively low. This means that terrain undulation and surface roughness are the main factors leading to landslides, and mountain terrain with high terrain undulation and surface roughness tends to be more likely to cause landslides, while landslides with low terrain undulation and surface roughness have a lower probability of landslide potential. The comprehensive analysis concluded that terrain relief and surface roughness were identified as the main factors in the assessment indicators, which are closely related to the degree of landslide damage. Therefore, special attention needs to be paid to the changes in these geomorphic features. This study provides a scientific basis for landslide prevention and ecological restoration in the Helan Mountain area.**

Keywords- landslide; digital elevation model; comprehensive evaluation of principal component analysis; cluster analysis

I. INTRODUCTION

Mountainous regions are among the areas prone to natural disasters[1]. Among them, landslides are a common geological hazard[2]. Landslides not only pose significant challenges to the ecological environment but also threaten human activities[3]. To determine the primary factors influencing landslides, the selection of influencing factors is crucial. Efficient evaluation factors can improve the accuracy of the model and make the experimental results more accurate. In-depth research on the impact of different influencing factors on landslides not only helps to understand the restoration process of mountain ecosystem, but also provides scientific basis for regional disaster prevention and reduction[4].

Helan Mountain is located at the junction of the Ningxia Hui Autonomous Region and Inner Mongolia Autonomous Region. It is an important ecological barrier in the Yellow River Basin. Its rich mineral resources and vegetation diversity make it a hotspot for ecosystem protection. However, due to the influence of human activities, mining and natural factors, the frequency of landslides is relatively high, and the ecological environment of Helan Mountain is faced with serious challenges[5].

II. STUDY AREA PROFILES AND DATA SOURCES

A. Research significance

The Helan Mountain region has attracted much attention from researchers because of its complex topography, diversified ecosystem and unique natural landscape. Over the past few decades, the ecology of Helan Mountain has undergone tremendous changes[6], which is closely related to the changes in human activities and the natural environment. Vegetates is an important part of the mountain ecosystem, which plays an important role in maintaining ecological balance and protecting the soil and water source. Helan Mountain is a landslide-prone area, and landslide disasters have a serious impact on the local social economy and people's life. Landslides will destroy the local ecological environment, lead to soil erosion, soil erosion and other problems[7], and then affect water quality and biodiversity. Therefore, in-depth research on ecological restoration and reconstruction of Helan Mountain and understanding of the impact of different types of impact factors on landslide[8] can provide scientific basis for ecological protection and restoration of Helan Mountain and similar areas. The flowchart of this paper is shown in Figure1:

2023 2nd International Conference on Machine Learning, Cloud Computing and Intelligent Mining (MLCCIM)

979-8-3503-2853-0/23/$31.00 ©2023 IEEE
DOI 10.1109/MLCCIM60412.2023.00083

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.

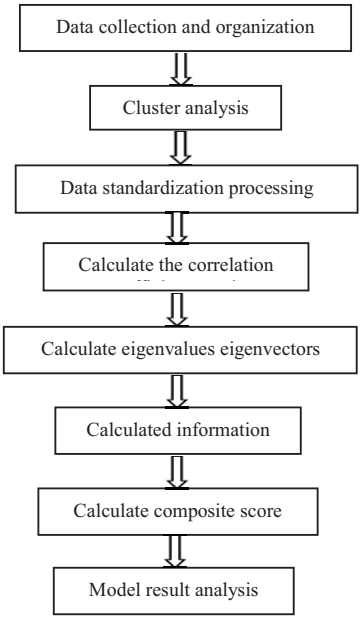


Figure 1. Flow chart.

B. Geological condition

Helan Mountain is one of the edge mountains of the Loess Plateau in the northern China. The average altitude is between 2000-3000 meters. The Apoba is the highest peak of the mountain. The altitude is about 3556 meters. It is affected by both humid monsoon climate and arid continental climate. The study area is located between 105°25' to 106°12' east longitude and 37°38' to 38°50' north latitude, with an approximate width of 16 to 25 km from east to west and a length of about 250 km from north to south. The total area is approximately over 7100 square kilometers, spanning a significant north-south distance[9]. The terrain and geomorphology are complex, including steep slopes, canyons and plateaux, with strong internal and external dynamic geological processes. Therefore, soil lithology is also complex and diverse. The 56 landslide points in Helan Mountain are shown in Figure 2:

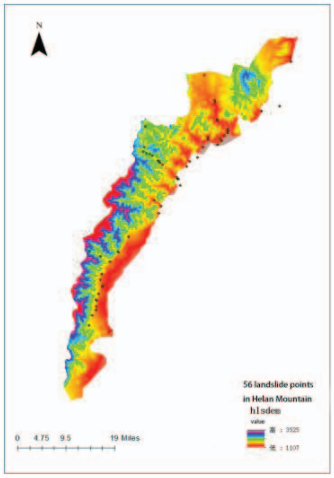


Figure 2. Map of 56 landslide sites in Helan Mountain

C. Soil lithology

Helan Mountain is a north-south mountain range, and its structural features are mainly influenced by crustal movements. Its landforms are diverse, including hills, canyons, platforms and cliffs[10]. Due to the action of crustal movement, there are many fault blocks and fault basins in Helan Mountain region. The mountains are mainly composed of metamorphic rocks, sedimentary rocks and volcanic rocks. Among them, metamorphic rocks include tingling rock, Yunmu film hemp and thousands of rocks. The sedimentary rocks include sandstone, shale and width. The volcanic rocks include Xuanwu Rock and Anshan Rock. These rocks formed the sedimentary rocks and Huosheng rocks from the ancient generations to the third period, giving rise to a diverse rock landscape in the region.

The soil types in Helan Mountain mainly include loess, rocky desertification soil and mountain brown soil. Loess is the most common soil type. Rocky desertification soil is formed due to the large relief of terrain and serious soil erosion. Mountain brown soil is mainly distributed in the foothills and foothills. The region is rich in coal, iron ore, copper, aluminum, tin and other mineral resources; Common mineral components in soil are quartz, feldspar, mica and so on. The acidity and alkalinity of soil vary, there are acidic soil, there are neutral and alkaline soil. Due to the influence of topography and soil erosion, the soil fertility of Helan Mountain region is different, the soil in some areas is fertile, while the soil in others is poor. The actual scene of Helan Mountain field survey is shown in Figure 3:



Figure 3. Helan Mountain landslide real picture.

D. Source of data

A variety of data sources were used in this study. Remote sensing image analysis was combined with field investigation, and field investigation was conducted through field visits and sample surveys. 56 landslide points in Helan Mountain area were investigated, including landslide location, scale, characteristics and other information. The remote sensing image mainly uses high-resolution remote sensing image data in 2021[11], and the data sources are shown in Table I:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.

TABLE I. DATA SOURCE.

Data	Source
15 m DEM	Geospatial data cloud platform
Helan Mountain landslide point data	Ningxia Hui Autonomous Region Land and Resources Survey and Monitoring Institute.
30m land use type data	Data set in The 30m annual land cover and its dynamics in China from 1990 to 2019, co-authored by two professors Yang Jie and Huang Xin from Wuhan University, which has been updated to 1985-2020

III. LANDSLIDE SUSCEPTIBILITY ASSESSMENT

Landslide refers to the phenomenon of downward movement of loose material (such as soil, rock, etc.) on the surface or hillside under the action of gravity. It usually occurs in areas such as steep slopes, slopes, earthquakes and heavy rainfall. The occurrence of landslide will not only bring a series of disasters such as land destruction, house collapse and road interruption, but also cause great harm to human beings and the environment[12]. Figure 4 shows a schematic diagram of the landslide.

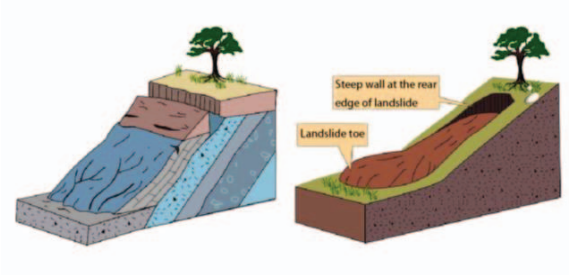


Figure 4. Landslide diagram.

A. Cluster analysis

The formation of mountain landslides is influenced by many factors, including the development degree of the landslide, soil and rock properties, geological structures, and human activities. Based on existing research findings, we utilized data from landslides in Helan Mountain and employed cluster analysis to classify landslides caused by different influencing factors. We preliminarily identified eight potential influencing factors[13]: slope angle, aspect, curvature, planar curvature, profile curvature, terrain relief, surface roughness, and terrain wetness index. Among them, slope angle, aspect, curvature, and terrain wetness index can be considered continuous data, while planar curvature, profile curvature, terrain relief, and surface roughness belong to discrete data. To achieve consistent data types and account for differences among categories, we discretized the continuous data into five intervals (labeled as X1-X8 in Table II).

TABLE II. SUMMARY OF IMPACT FACTORS.

Influence factor	Category
Degree of slope	≤6. 5、(6. 5, 13. 7]、(13. 7, 21. 1]、(21. 1, 30. 1]、>30. 1
Aspect of slope	Plane, North, northeast, East, southeast, South, southwest, west, northwest
Curvature	≤-0. 64、(-0. 64, -0. 18]、(-0. 18, 0. 21]、(0. 21, 0. 7]、>0. 7
Plane curvature	≤-0. 64、(-0. 64, -0. 18]、(-0. 18, 0. 21]、(0. 21, 0. 7]、>0. 7
Profile curvature	≤-0. 88、(-0. 88, -0. 25]、(-0. 25, 0. 21]、(0. 21, 0. 88]、>0. 88
Degree of relief	≤1、(2, 3. 4]、(5. 6, 7. 8]、(9. 10, 11. 12]、>12
Surface roughness	≤1. 03、(1. 03, 1. 09]、(1. 09, 1. 17]、(1. 17, 1. 3]、>1. 3
Topographic humidity index	≤2、(2, 3. 6]、(3. 6, 5. 8]、(5. 8, 8. 8]、>8. 8

In order to evaluate the correlation between these eight variables[14], Pearson correlation coefficient was calculated using SPSS software. From the data obtained in Table III, it can be found that the correlation coefficient between slope, topographic relief and surface roughness is less than 0.3; The correlation coefficient between curvature, plane curvature and profile curvature is less than 0.5. The correlation coefficient between surface roughness and relief is less than 0.3. The results showed that there was no significant correlation among the factors. Therefore, there is no substitution between these indicators, and none of them should be ignored.

TABLE III. ANALYSIS OF INFLUENCE FACTOR CORRELATION.

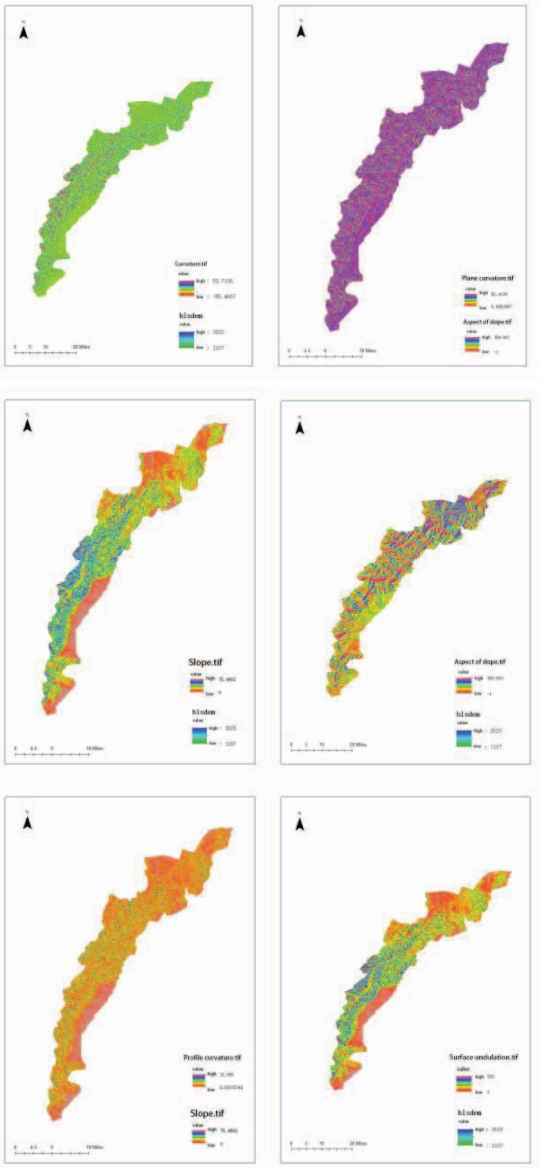
Factor	Degree of slope	Aspect of slope	Curvature	Plane curvature	Profile curvature	Degree of relief	Surface roughness	Topographic humidity index
Degree of slope	1.00							
Aspect of slope	0.02	1.00						
Curvature	0.01	0	1.00					
Plane curvature	0.01	0	0.40	1.00				
Profile curvature	0.01	0	0.47	0.39	1.00			
Degree of relief	0.22	0.03	0.01	0.01	0.01	1.00		
Surface roughness	0.20	0.03	0	0.02	0.10	0.28	1.00	
Topographic humidity index	0.25	0.10	0.10	0.08	0.31	0.04	0.23	1.00

B. Comprehensive evaluation

The high level data of Ningxia Province and Inner Mongolia Province were downloaded from the geospatial data cloud, spliced and clipped by ArcGIS software, and a digital elevation model (DEM) was obtained. The elevation range was 1107~3525m, with a relatively large elevation span, low in the south and high in the north, and the mountain landscape in the south was generally flat. Next, we adopted the SVM-LSM toolbox based on Baidu

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.

PaddlePaddle frame training. This toolbox is a landslide susceptibility mapping toolbox based on surface data of support vector machine, using this tool can calculate the slope, slope, curvature, plane curvature, sectional curvature, terrain undulating, surface roughness and terrain humidity index. Figure 5 shows the results of the calculation.



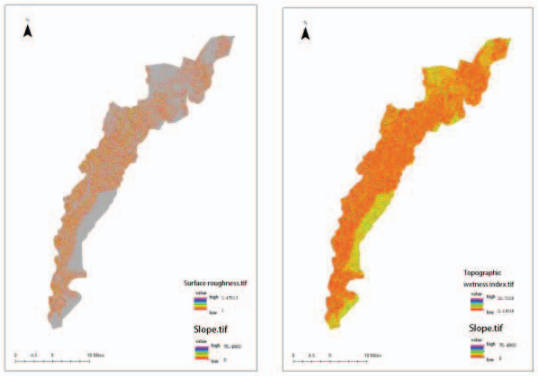


Figure 5. Landslide impact factor layer.

(1) The principal component analysis comprehensive evaluation method was adopted to carry out weight calculation and comprehensive evaluation of 8 indicators. First, the 8 indicators were regarded as different subjects, and then the original data were standardized. Formula (1) was used to convert each indicator X_{ij} into a standardized indicator $\tilde{x}_{ij}$, and the coefficient matrix was established, as shown in Figure 6:

$$\widetilde{a}_{ij} = \frac{a_{ij} - \mu_j}{s_j}, i = 1,2,\cdots,n; j = 1,2,\cdots,m; \tag{1}$$

Formula:

$$\mu_j = \frac{1}{n}\sum_{i=1}^{n} a_{ij}; s_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(a_{ij} - \mu_j)^2}, j = 1,2,\cdots,m, \tag{2}$$

That is, between μ_j, s_j and s_j is the original mean and sample standard deviation of the j index.

Correspondingly, said:

$$\widetilde{x}_j = \frac{x_j - \mu_j}{s_j}, j = 1,2,\cdots,m \tag{3}$$

Is a standardized indicator variable.

(2) Calculate the correlation coefficient matrix R. The correlation coefficient matrix $R = (r_{ij})_{m\times m}$, is:

$$r_{ij} = \frac{\sum_{k=1}^{n} \tilde{a}_{ki} * \tilde{a}_{kj}}{n-1}, i, j, \cdots, m, \tag{4}$$

Where: $r_{ii} = 1, r_{ij} = r_{jr}$, , are the correlation coefficients of the i index and the j index.

(3) Calculate eigenvalues and eigenvectors:

$$y_1 = u_{11}\widetilde{x}_1 + u_{21}\widetilde{x}_2 + \cdots + u_{m1}\widetilde{x}_m,$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.

$$y_2 = u_{12}\tilde{x}_1 + u_{22}\tilde{x}_2 + \cdots + u_{m2}\tilde{x}_m,$$
$$\vdots$$
$$y_m = u_{1m}\tilde{x}_1 + u_{2m}\tilde{x}_2 + \cdots + u_{mm}\tilde{x}_m, \quad (5)$$

(4) Calculate the information contribution rate of eigenvalues:

$$b_j = \frac{\lambda_j}{\sum_{k=1}^{m}\lambda_k}, j = 1,2,\cdots,m \quad (6)$$

(5) Calculate the comprehensive score and evaluate according to the comprehensive score:

$$Z = \sum_{j=1}^{p} b_j y_j \quad (7)$$

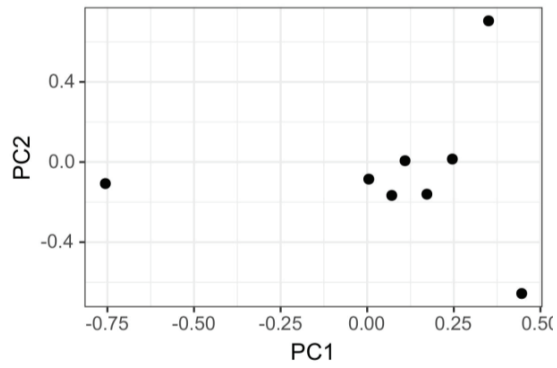


Figure 6. Diagram of principal component analysis results.

TABLE IV. PRINCIPAL COMPONENT ANALYSIS RESULTS.

Serial number	Characteristic root	Contribution rate	Cumulative contribution rate
1	4.8239	68.9135	98.9135
2	1.6227	23.1816	92.0950
3	0.3146	4.4948	96.5898
4	0.2265	3.2363	99.8261
5	0.0116	0.1652	99.9913
6	0.0006	0.0087	100.0000
7	0.0000	0.0000	100.0000
8	0.0000	0.0000	100.0000

The results show that the cumulative contribution rates of the first and second eigenvalues are both above 90%, indicating the effectiveness of PCA. Based on this, four principal components (with a cumulative contribution rate exceeding 98%) were selected for comprehensive evaluation. Table V shows the eigenvectors associated with the four eigenvalues.

TABLE V. PRINCIPAL COMPONENT ANALYSIS RESULTS.

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
1	0.1214	0.6885	0.7086	0.0947	0.0345	0.0018	0.3245	0.5196
2	0.0785	0.7093	0.7006	0.0177	0.0022	0.0004	0.4562	0.3451
3	0.4367	0.0875	0.0534	0.5414	0.1311	0.6989	0.2367	0.1593
4	0.4354	0.0966	0.0632	0.5501	0.1529	0.6864	0.5462	0.6429

Thus, the four principal components are:

$$y_1=0.3497x_1+0.359x_2+..+0.2452x_{10}$$
$$y_2=-0.1972x_1+0.0343x_2+\ldots-0.286x_{10}$$
$$y_3=-0.1639x_1-0.1084x_2+\ldots+0.8637x_{10}$$
$$y4=-0.1022x_1-0.2266x_2+\ldots-0.2457x_{10} \quad (8)$$

From the coefficients of the principal components, it can be inferred that the first principal component reveals the characteristics of slope, the second principal component reveals the characteristics of slope direction, the third principal component mainly reveals the characteristics of curvature, and the fourth principal component mainly reveals the characteristics of plane curvature. The values of the four principal components were derived by bringing the normalized data of the original eight indicators into the expressions of the four principal components. The contribution rates of the four principal components were utilized as weights to establish a comprehensive evaluation model of principal component analysis.

$$Z=0.7502y_1+0.1577y_2+0.0536y_3+0.0206y_4 \quad (9)$$

By substituting the values of the four principal components into the above formula, the comprehensive evaluation value can be calculated and sorted according to the calculation results. The specific sorting results are shown in Table VI.

TABLE VI. EVALUATION INDEX WEIGHT RANKING TABLE

Influence factor	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
Comprehensive valuation	15.278	4.227	1.309	1.201	1.007	0.926	0.849	0.755
weight	3	6	7	4	5	1	2	8

C. *Result presentation*

By comparing the remote sensing data and ground survey data of the Helan Mountain area with the comprehensive ranking results in the table above, and comparing the actual values of the 8 indicators in Table 1, we can conclude that the slope obtained under this model has a greater impact on landslides. Therefore, it is necessary to strengthen the monitoring and prevention of geological disasters in the Helan Mountain area, reduce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.

their negative impact on the ecological environment, and focus on areas with larger slopes. After a landslide disaster occurs, effective soil restoration work needs to be carried out to promote ecological restoration.

IV. Conclusion

Based on the evaluation of landslide susceptibility of Helan Mountain with different evaluation factor combinations, the optimal evaluation factor combinations and evaluation models are selected. Through cluster analysis, the classification modeling of landslide factors is realized. Through the analysis of the main component, the cumulative contribution rate of the characteristic roots is more than 90%, which proves that the reliability and accuracy of this method has a certain practical value determined that the ups and downs of terrain and the roughness of the surface in Helan Mountain areas landslide landslides The importance. In areas with large slopes and poor soil types, the landslides are more serious. Therefore, when the implementation of landslides and ecological restoration measures for mountain landslides, special attention should be paid to the changes in the terrain and surface roughness of the Helan Mountain area, focusing on changes in the characteristics of the landform characteristics of the prone to landslide area, and taking corresponding measures to reduce the destruction and destruction and Personnel security risk.

Acknowledgment

This work supported by The Key Research and Development Program Project of Ningxia (2021BEG03001), The 2023 Fujian Provincial Foreign Cooperation Science and Technology Plan Project (2023I0047), The 2022 Longyan Industry-University-Research Joint Innovation Project (2022LYF18001), The 2023 Fujian Natural Resources Science and Technology Innovation Project (KY-060000-04-2023-2002).

References

[1] B Chen, Z Li, W Huang, et al. Spatial distribution and influencing factors of geological hazards induced by the 2022 Luding M_w6.6 earthquake in Sichuan Province[J]. Journal of Earth Sciences and Environment,,2022,44(06):pp. 971-985.

[2] Z Li, C Zhang, B Chen, et al. A technical framework for landslide prevention based on multi-source remote sensing and its engineering application[J]. Earth science, 2022, 47(06):pp. 1901-1916.

[3] M Li, G Li. Effects of vegetation types on tree-ring reconstruction of NDVI in Helan Mountain[J]. Journal of Gansu Agricultural University, 2020,55(05):pp. 152-161.

[4] H Zhu. Risk assessment and disaster mechanism of debris flow at the eastern foot of Helan Mountain[D]. East China University of Technology, 2021.

[5] Y Lin, X Nan, Z Hu, et al. Temporal and spatial changes of vegetation coverage and its ecological security assessment in typical ecologically fragile areas in Northwest China: A case study of Helan Mountain in Ningxia[J]. Journal of Ecology and Rural Environment, 2022, 38(05):pp. 599-608. (in Chinese)

[6] Z Zhou, L Dong. Investigation on the loss and restoration of garden trees in Xiamen after a strong typhoon -- taking Huaqiao University Xiamen Campus as an example[J]. Landscape architecture, 2018,25(06):pp. 41-46.

[7] K Yang, X Ji, L Mao, et al. Comparison of numerical simulation and topographic effects of two extremely heavy flood-causing rainstorms in Helan Mountain[J]. Arid weather, 2020, 38(04):pp. 581-590.

[8] X Ma, X Liu, X Gao, et al. Beidou helps build intelligent robot solutions for building surface care [J]. Wireless interconnection technology, 2020, 17(17):pp. 56-57. (in Chinese)

[9] Y Cui, Z Chang, H Zuo, et al. Study on risk and impact regionalization of mountain flood disaster in Yinchuan section of Helan Mountain during different recurrence periods[J]. Arid region geography, 2020,43 (04):pp. 859-870.

[10] G Wang, Z Qiang, C Cao, et al. Evaluation of Karst ground collapse susceptibility based on geographic detector and analytic Hierarchy process: A case study of Zhongliangshan area in Chongqing[J]. Karst in China, 2022, 41(01):pp. 79-87.

[11] Y Dong, M Ding, X Li, et al. Evolution process of 2018 Baige landslide in Jinsha River Basin based on optical remote sensing pixel offset[J]. Journal of Earth Sciences and Environment, 2022,44(06):pp. 1002-1015.

[12] W Huang, M Ding, D Wang, et al, Evaluation of landslide susceptibility along Sichuan-Tibet traffic corridor based on layer adaptive weighted convolutional neural network[J]. Earth science, 2022, 47(06):pp. 2015-2030. (in Chinese)

[13] S Frizzi, R Kaabi, M Bouchouicha, et al. Convolutional neural network for video fire and smoke detection [C]. In IECON 2016-42nd Annual Conference of the IEEE Industrial Electronics Society, 2016:pp. 877-882.

[14] S Wang, J Zhuang, J Zheng, et al. Evaluation of landslide susceptibility of Kangding-Litang section of CZ railway based on deep learning[J]. Journal of engineering geology, 2022,30(03):pp. 908-919.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:42:29 UTC from IEEE Xplore. Restrictions apply.